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Climate actions on the neighbourhood level—Individual, collective, cultural, and socio-structural factors

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Abstract

This paper takes a multi-perspective approach to understand drivers and barriers of climate action on the neighbourhood level. We start with the assumption that climate actions on the level of citizens are most motivating and promising, when conducted jointly within established social systems like neighbourhoods. A survey implemented in neighbourhoods (3 in Austria, 2 in Norway, 2 in Italy, 2 in Finland). The neighbourhoods were partly in rural communities (4) and partly in urban or semi-urban areas (5). In total, 1.084 answers were retained between summer 2022 and summer 2023. The impact of factors from the different perspectives on the self-reported number of implemented climate actions were tested in a stepwise structural-equation-modelling-approach. The analyses show that intentions to act both on the individual and collective level impact climate actions as represented by behaviour in four domains (travel, diet, protest, and general climate action) implemented by citizens in the neighbourhoods, but individual intentions are more important. In addition, local cultural aspects have an impact on climate action, as indicated by the two extremely rural Finnish neighbourhoods being different on many variables. On the socio-structural level, males and households with younger children report less climate action, whereas larger households in general and people with university degree report more. Intentions to act individually are mostly determined by perceived individual efficacy and attitudes, but also selected cultural and socio-structural factors. Collective intentions to act depend on the social capital in the neighbourhood, collective efficacy, and social norms, as well as selected socio-structural and cultural factors. Concluding, this paper emphasises that in order to understand and stimulate climate-related action of citizens, the individual, collective, cultural and socio-structural factors must be taken into account and that the level of neighbourhoods, where everyday action takes place, is a relevant unit of analysis to do so.

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Introduction

Environmental crises like climate change and biodiversity loss require fast global action [1, 2]. While the fundamental decisions need to be taken on the inter-government level, a strong focus has also been put on the role of the individual in this transition—especially in western, resource intensive societies [3]. At the same time, this approach has been criticized for over-emphasising the individual's role and "blaming the consumer", thereby ignoring the role of systematic socio-structural and cultural influences [4–6]. Alternative approaches propose that individuals are strongly embedded in socio-technical systems, which means that there are both socio-cultural, political, as well as technological systems that shape their choices in parallel to individual factors such as for example attitudes, knowledge, or self-efficacy [7–9].

Substantial research has been conducted on many of the aforementioned levels: On the individual level, many studies in the last decades have addressed drivers of people's climate relevant actions, mainly from a psychological or micro-economic perspective. Gifford [10] for example presents a compelling overview of psychological barriers towards climate action, including factors such as perceived risks, cognitive biases, or ideologies. In a recent review, Steg [11] summarizes research on the role of different value orientations, emotions, social influence, as well as barriers that interfere with acting in line with one's values to protect the climate. Swim et al. [3] very comprehensively present the psychological state-of-the-art with respect to psychological effects of climate change, but also drivers of climate action. Whereas psychologists and micro-economists have a focus on the micro perspective, sociologists often focus on the effects of the larger social and economic structures on peoples' climate actions, on questions of power and power differences, as well as political systems and structures (for an overview, see for example [12]), with that navigating mostly the meso and macro level. However, also these analyses often implicitly look at how individuals act in response to or limited by these higher-level structures.

Recently, a shift in perspective from the individual to collective climate action has occurred. On the one hand, this has been driven by work in the practice-theoretical approach, which understands human effect on climate as the result of practices which again are formed by the interplay of material infrastructures, governance, resources, and meaning. Sahakian and Wilhite [13], for example propose practice theory as a fruitful lens through which to analyse change of consumption practices, as it distributes the agency between people, the material things, and the social context, which gives a more wholistic understanding of where the drivers and barriers are created. Hargreaves [14] used this approach to study how behaviour change at a workplace towards more pro-environmental practices occurred. He found that this perspective helped to understand the implicit barriers of established practices. On the other hand, also psychological research has to a growing degree shifted to the role of collective rather than individual action, often driven by an urge to understand climate protest movements. Bamberg et al. [15, 16] for example argue that collective action is to a stronger degree motivating than individual action. Barth et al. [17] propose a framework model for collective environmental action, where both individual and collective elements variables play important roles, and the "ingroup" (hence the group one feels a belonging to) becomes a central concept. This draws attention to where to find such "natural" groups to study collective climate action in. Hargreaves [14] used a workplace as an arena of collective action, with established relations and practices. In our study, we propose neighbourhoods as a similar arena for climate action.

Neighbourhoods are where people live and where they spend most of their free time and where many concrete climate actions individuals can engage in are implemented [18]. Compared to other arenas, such as workplaces, universities, student dorms, and the like, neighbourhoods are relatively understudied. Joshi et al. [19] recently strongly argued for

neighbourhoods as a logical unit for bottom-up climate action, but they also identify challenges of stimulating collective action on this level, such as limited participation, small-scale impacts (which might be demotivating), and limited resources. Yet, a comprehensive study analysing and comparing the drivers and barriers of climate action on the neighbourhood level in a selection of European neighbourhoods with a rich variety of characteristics, has to our knowledge not been conducted before. Such a study can help unravelling the complex social, psychological, political, and structural conditions of local climate action. Thus, the aim of this paper is to analyse and better understand how individual, collective, cultural, and socio-structural factors contribute to the development of local climate-action, when investigated simultaneously.

A comprehensive multi-level perspective on local climate action

In this article, we define the *neighbourhood* according to the perspective of the ecological theory of Bronfenbrenner [20], i.e. as a *microsystem*. A neighbourhood is essentially a unit representing a group of people living together in proximity, forming a social system of interrelations. A definition of the complex phenomenon of "neighbourhood" as presented in Carrus et al. [21] includes the physical dimension [22, 23], sociodemographic characteristics [24], aspects of identity [25], institutional and administrative aspects [26], as well as social relationships [22]. However, before we explore the relation of neighbourhoods to climate actions more, we first need to understand, what constitutes a neighbourhood for its members.

In a recent paper, von Stülpnagel, Brand, and Seemann [27] delineate the physical space people assign to their understanding of a neighbourhood and find that the prevailing approach of assuming a neighbourhood being defined purely by distance to the residence of the respondent falls substantially short. They rather recommend using an approach based on cognitive mapping of the neighbourhood, which also allows respondents to take other factors into account. A neighbourhood can be defined as a physical space with boundaries given by waterways, main streets, by administrative references or by a particular type of social relations. Residents of a given neighbourhood may identify its boundaries differently than administrative boundaries by referring to social or cultural criteria [22]. In our study, we used a combination of physical-structural borders (e.g., groups of residential buildings, dividing traffic infrastructures), administrative units, as well as a social definition of the neighbourhoods (by probing residents for what they understand as their neighbourhood in an early stage of the project) as our starting point for the analyses.

The physical aspects of neighbourhoods are considered predictors of residential satisfaction or of the perception of urban (in)security [28]. These considerations relate to the structural and cultural dimensions of life in neighbourhoods (see Fig 1 below, which gives an overview about the different perspectives we see as relevant for climate action on the neighbourhood level). In addition to its physical and cultural dimensions, a microsystem is also an integrated system of activities, "roles, and interpersonal relations experienced by the developing person in a given setting with particular physical and material characteristics" [20]. Moving towards the more institutional dimensions, we can observe that the neighbourhood is the place where a lot of everyday face-to-face communication takes place: Everyday problems are discussed, especially if connected to the neighbourhood or if it affects multiple persons in this neighbourhood. It is also very likely that opinions on various subjects are exchanged in this everyday communication.

The development of technologies has a primary role in what people can do both with respect to communication but also other practices. For example, the way relations between citizens of the same neighbourhoods unfold changed substantially through development

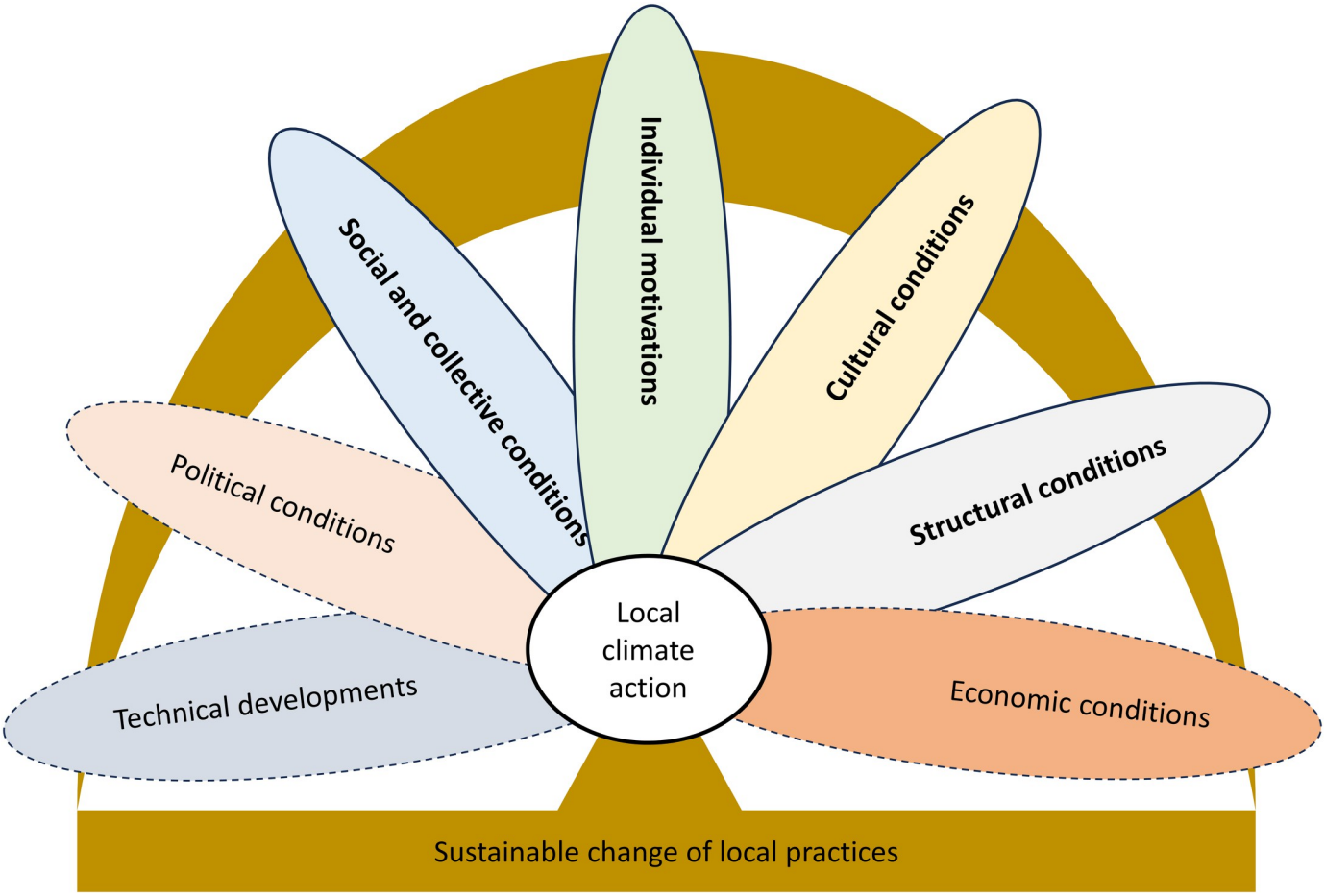


Fig 1. An overview of the multi-perspective model of local climate action (technology, policy, and economy are not studied in this paper).

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communication technology; if before we used to go knocking to ask for some salt or a screwdriver, now it is enough contact your neighbour via social media or to visit an e-commerce site to order what is needed to be delivered home in half a day. The social interaction is completed with the support between the members of the society and the networks in which each member is inserted. Social support mainly refers to the availability of the other as a resource, to the presence of a support that is given by the co-presence of others similar to me [29].

In this way, the different levels are always inevitably linked to each other. Individual attitudes are interdependent with social and economic conditions, it is difficult to think of the well-being of the planet if one does not live in a condition of sufficient personal well-being. Structural conditions shape what people can achieve in a given neighbourhood together and which topics are prevalent for them. The larger technical developments (like communication technology, energy efficiency, transport technology) and political conditions (laws, city development, etc.) are defining the space in which people can unfold their behaviours and practices individually and collectively. Aspects of individual and social well-being predispose people to pro-environmental motivations that can be supported, fostered, and carried out by institutions and politics.

While intertwined with each other, the design of our study puts limits on to which degree we can unravel the seven proposed layers of local neighbourhood climate action. With the

relatively small number of (all European) countries (four) and neighbourhoods (nine) we had the opportunity to study, the differences with respect to political and economic conditions, as well as broader technological trends are limited and confounded with other higher level cultural factors. We therefore decided to not focus on these factors but rather approximate them jointly by testing country effects. This does not mean, however, that we do not consider them relevant, and we would like to study them in more detail in future studies with a much higher number of neighbourhoods from a larger and more diverse set of countries, which would also allow for a (in a statistical sense) multi-level approach (see also the discussion section). In the following sections, we zoom in on the four aspects that we could study in more detail to provide the ground for our analysis in this paper.

Individual factors influencing climate action

We start our exploration of potentially impactful factors with the individual level. Environmental psychology and environmental sociology have produced a multiplicity of theoretical frameworks [30], but in particular, the link between attitudes, norms, self-efficacy and actions was examined (e.g., [31]).

Consistently, intentions to act have been linked to climate-related actions, although this link is fragile and can be disrupted if strong habits or situational barriers are interfering [32]. These intentions are strongly depending on the attitudes a person has towards the different behaviours, thus, the general evaluation of the behavioural alternatives as positive or negative. Furthermore, research shows that social norms (social influence) is an important factor, particularly in close-knit social systems such as neighbourhoods [33]. Also, an assessment of one's capability to implement the behaviour, often referred to as self-efficacy, is among the most consistent factors that influence taking climate action [34]. We are aware that there are many more individual-level variables which have been linked to climate action (e.g., habits, personal norms, knowledge), but we decided to stick to the ones included in the most applied single theory (Theory of Planned Behaviour [31]) to be able to cover a substantial number of factors from the other areas as well in the analysis. Therefore, we have the following hypotheses for the relation in the individual factors:

H1a: Individual intentions to act against climate change result in a higher level of (self-reported) climate action.

H1b: Higher perceived individual efficacy results in stronger individual intentions to act against climate change.

H1c: Stronger pro-climate attitudes result in stronger individual intentions to act against climate change.

Furthermore, we expect that social norms rather have an influence on collective intentions than individual intentions (see next paragraph).

Social and collective factors influencing climate action

Social norms can be considered a linking factor between the individual and the collective level, and therefore be relevant on both levels, but we expect their influence to mainly manifest on the collective level. Bamberg, Rees, and Seebauer [15] proposed that such collective level factors can become relevant for collective climate action, adding collective intentions to act and collective efficacy to the list of factors that might be relevant. For our analysis, we will therefore make a distinction between individual level factors (in our case individual intentions, attitudes,

self-efficacy, and social norms), and collective, neighbourhood-level factors, which mirror these.

Social capital in the neighbourhood has been studied for a long time as a driver of collective actions on the local level. Social capital has been originally defined from an economic perspective as the capital that exists in the relations between people [35]. Adopted by social scientists, neighbourhood social capital has been further refined as including concepts such as social networks, norms of reciprocity, ability for collective actions, shared behavioural patterns, and social norms [36]. In our study, we understand social capital as the norms and collective resources to act, collective creativity, and relation between the members of the neighbourhood. Higher social capital has been linked to better health and health behaviour [36], more liberal voting [37], better climate change adaptation [38], or more climate change mitigation efforts [39]. This short review results in the following hypotheses:

H2a: Collective intentions to act against climate change on the neighbourhood level result in a higher level of (self-reported) climate action.

H2b: Higher perceived collective efficacy results in stronger collective intentions to act against climate change.

H2c: Stronger social norms result in stronger collective intentions to act against climate change.

H2d: Higher social capital in the neighbourhood results in stronger collective intentions to act against climate change.

Cultural factors influencing climate action

Another level up in the analysis, the cultural level can be located. With respect to energy behaviour, the cultural influences have often been conceptualized as energy cultures (as for example in [40]). Energy cultures capture the cultural assumptions on which behaviours are adequate, the collectively shared experiences, narratives, beliefs, understandings, technologies, and activities. They shape the public perception and reaction to climate change [41]. Also, cultural worldviews influence the ways in which people perceive climate change risk [42–44]. Chan and Tam [45] found that the association between climate change concern and mitigation behaviour was particularly strong in societies where self-expression affordance had high levels, which are countries with lower threat for serious diseases, better governance, good economic development, and stronger individualism.

However, measuring cultural factors that may influence climate action is challenging. Whilst highly integrated frameworks such as "Energy Cultures" [40] provide orientation and can serve as a basis for specific adaptations, their empirical operationalization for statistical modelling is complex due to their large number of required variables. Therefore, using variables like the country of residence or urban vs. rural environment as proxies for "culture" is a common procedure [46]. For example, Thøgersen used "country" as proxy for the specific constellations of factors potentially influencing housing related energy saving practices and diets in different contexts [47, 48]. Similarly, Schwarzinger, Bird, and Skjølsvold [49] used "country" as cumulative proxy for the local cultural context in a study on multi-domain energy behaviour. We follow this approach in our study. Similarly, the rural-urban dimension represents a proxy for cultural differences. Zeigermann, Kammerer, and Böcher [50] show, for example, that public funding of climate change mitigation is less culturally anchored in rural areas in Germany than in urban areas. This short review results in the following hypotheses:

H3a: There are country differences in central variables like individual and collective intentions and climate action, as well as in the predictors of these. The direction of the impact is not determined a priori.

H3b: There are differences between rural and urban neighbourhoods in central variables like individual and collective intentions and climate action, as well as in the predictors of these. The direction of the impact is not determined a priori.

Socio-structural factors influencing climate action

The influence of socio-structural factors on climate behaviour and climate action has been widely reflected in research. In empirical work, recent findings emphasize the importance of a thorough coverage of socio-structural factors in the analysis of climate related behaviour and resulting impacts. For example, results obtained by Schweighart, Schwarzinger, and Bird [51] show that using household composition in regression models increases the explanatory power. Reichl et al. [52] show for example that people identifying as male, people of certain age groups (in working age) and people living in larger household show fewer climate actions, whereas the number of children of a respondent was positively related to climate actions. Furnham & Robinson [53] find that identifying male and being older were negatively related to self-reported climate change mitigation action, whereas higher education was positively related. Based on this literature, we expect the following relations:

H4a: Age has an influence on central variables like individual and collective intentions and climate action, as well as predictors of these. The direction of the impact is not determined a priori.

H4b: Participants who identify as males score lower on central variables like individual and collective intentions and climate action, as well as predictors of these.

H4c: Increased household size leads to lower scores on central variables like individual and collective intentions and climate action, as well as predictors of these.

H4d: Number of children bellow 14 years of age leads to higher scores on central variables like individual and collective intentions and climate action, as well as predictors of these.

H4e: Higher education leads to higher scores on central variables like individual and collective intentions and climate action, as well as predictors of these.

H4f: Working full time leads to lower scores on central variables like individual and collective intentions and climate action, as well as predictors of these.

Analysed factors from the multiple perspectives

As the literature review demonstrates, many factors on many different levels have been shown to impact how likely it is that citizens make decisions to engage in climate action in their local environment. Table 1 displays the specific factors from the four analysed levels that have been selected to be implemented in our study. To limit the length of the survey, the number of factors to be included had to be restricted. On the individual level, four factors from the Theory of Planned Behaviour have been selected: (individual) intentions to act, attitudes towards climate action, and perceived (individual) efficacy. Two of these are mirrored on the collective

Table 1. Mapping of the selected factors on the four analysed perspectives of local climate action.

Individual level:	Social / collective level:
• Individual intention to act • Attitudes towards climate action • Perceived individual efficacy	• Collective intention to act • Perceived collective efficacy • Social norms • Social capital in the neighbourhood
Cultural level:	**Socio-structural factors:**
• Country (as proxy for the local culture) • Rural (vs. semi/urban) neighbourhood (as proxy for urban vs. rural worldviews)	• Age • Gender • Household size • Number of household members under 14 years • Education • Work situation

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level (intention, efficacy), whereas social norms is placed on the collective rather than the individual level. Social capital in the neighbourhood was also measured.

On the cultural level, two variables which serve as proxies for (potentially) different local cultures are used: (a) the country the respondents live in, and (b) rural vs. (semi)urban.

On the structural level, a number of sociodemographic variables like gender, age, household size, number of household members below 14 years of age, education, and work situation were recorded. The measurement instruments for each of the factors are described in more detail in the method section below.

Method

Ethical statement

The study was approved by the following committees for research ethics: Norwegian Agency for Shared Services in Education and Research, SIKT (Ref.nr. 121957, approved on 22/04/2022), Ethical Committee of VTT (Statement code 8_2022, approved 30/06/2022); Roma Tre Ethics Commission (approved in commission meeting 15/02/2022). For the Austrian sample, an ethical clearance was not required for an anonymous paper-pencil survey as per ethical procedures of Joanneum Research. Participants gave informed consent to participate after being informed about their rights at the beginning of the survey by delivering the paper-pencil survey in Austria or clicking the "begin survey" button in the online surveys. No minors or people unable to give consent were included in the study.

Sample

From summer 2022 to summer 2023 (data collections were not conducted synchronized in all four countries: Austria 01/07/2022-15/09/2022; Italy 01/09/2022-20/05/2023; Norway 03/06/2022-30/09/2022; Finland 30/06/2022-10/08/2022), citizens in nine European neighbourhoods were asked to answer the questionnaire including measures for the factors outlined above. Neighbourhoods were selected due to the following criteria: (a) they should be located in the countries of the respective researchers involved in the project so that investigations could take place without language barriers and that the researchers were familiar with the cultural context, (b) they should together represent a good distribution of rural and urban areas, as well as a spread of socio-economic conditions, and (c) first contacts with the municipalities in these areas had

already been established. This allowed us to start with an already existing level of trust between the research team and key local actors. The selection of neighbourhoods was conducted with a focus on diversity of living conditions, not representativity for each country, let alone Europe.

Before designing and implementing the survey, the neighbourhoods had been studied with qualitative methods (document analyses, expert interviews, interviews with residents, focus groups) to develop an understanding of the prevalent challenges and topics that would need to be reflected in the quantitative survey. S3 Table presents a short summary of the conclusions from that work, a more extensive analysis is presented in the project report [54]. Data collection varied between the neighbourhoods based on what the local research teams assessed as being the potentially most successful approach. In Austria, data was collected with paper-pencil questionnaires distributed in the selected neighbourhoods by mail as well as with an online questionnaire, also available in English and Turkish. In Norway, the researchers hired local adolescents to go from door to door in the neighbourhoods to distribute an invitation letter with the link to an online version of the questionnaire. In Italy, a similar approach was chosen. In Finland, a survey company was contracted to conduct the survey as a telephone interview with citizens in the selected regions.

In total, 1.084 responses were collected, distributed very unevenly across the neighbourhoods (see Table 2) due to two factors: the target neighbourhoods have very different sizes, varying from under 400 residents in the smallest to about 30,000 residents in the largest; recruitment methods differed in success rate. The neighbourhoods were also very different in their social profiles, which was the aim of the study design: Some have an older population with many retired people, some have a younger population with many families with children. In some, more men answer than in others. There are differences in self-reported social status and education levels. All socio-demographics in Table 2 show statistically significant differences between the neighbourhoods (with $p \leq .002$).

Analysis strategy

The hypothized relations between variables were specified as a structural equation model. The model was estimated in MPLUS with a WLSMV estimator. First, a measurement model was tested for all latent variables (variables with more than one indicator question in the survey). Questions with cross-loadings or too small standardized loadings (below.50) were removed from the model. Measurement invariance across the four countries could not be tested due to small number of cases in some of the countries, however, equal loadings were enforced in the model across all countries. Second, a full model was specified with all assumed relationships included. Consecutively, all non-significant relations were removed from the model to reduce model complexity. Finally, modification indices were used to further improve model fit by adding three paths to the model which were not foreseen (a direct path from social capital to climate action, a direct path from attitudes to climate action, and a residual covariation between INDEFF3 and COEFF1, two items with a similar focus). We are aware that model modification removing non-significant paths and using modification indices is fitting the resulting final model to the dataset at hand. However, given the exploratory nature of the paper, we accept this risk and recommend for future studies to replicate the results. In this paper, only the final resulting model is presented (Fig 3 and Table 3), but intermediate models can be obtained from the corresponding author upon request.

Measurement instruments

Behaviour. To measure the generalized climate actions the respondents implement, they were confronted with a list of behaviours (see S1 Table) and were asked to indicate, which

Table 2. Overview of the respondents in the different neighbourhoods.

	Neighbourhoods									Total
	Admont	Eggenberg	Jakomini	Driva	Myrsletta	Simo	Pyhäntä	Guidonia	Macomer	
Country	Austria	Austria	Austria	Norway	Norway	Finland	Finland	Italy	Italy	
Type	Rural	Urban	Urban	Rural	Semi urban	Rural	Rural	Semi urban	Semi urban	
N	274	207	287	14	28	100	100	37	37	1.084
Percentage of total residents in the neighbourhood	5.5%	1.0%	1.0%	3.5%	5.6%	3.4%	6.3%	0.2%	0.4%	1.3%
18–34 years	15,7%	28,0%	36,2%		21,4%	8%	10%	22%	41%	23.2%
35–49 years	21,2%	27,5%	20,2%	30,8%	50,0%	9%	13%	38%	14%	21.4%
50–65 years	33,9%	25,1%	25,4%	46,2%	14,3%	26%	32%	41%	43%	29.3%
more than 65 years	29,2%	18,8%	17,8%	23,1%	14,3%	57%	45%	0%	3%	25.9%
Male	50,5%	38,3%	49,5%	50,0%	42,9%	59,0%	58,0%	38,90%	21,60%	47.8%
Female	49,1%	59,2%	50,2%	50,0%	53,6%	41,0%	42,0%	61,1%	78,4%	51.4%
Divers	0,4%	2,4%	0,3%	-	3,6%	-	-	-	-	
Average household size	2.1	2.0	1.9	2.9	3.3	2.1	2.3	2.4	2.7	2.2
Average number below 14 years	0.4	0.5	0.4	0.5	1.2	0.2	0.3	0.7	0.4	0.4
Social Status (0 = worst, 10 = best)	6.1	6.4	6.3	7.1	6.5	a	A	7.1	6.9	6.3
Primary School	4,4%	6,4%	7,5%	-	3,6%	18,0%	19,2%	5,40%	2,70%	8.2%
Vocational education	46,9%	20,8%	20,4%	14,3%	10,7%	48,0%	42,4%	2,70%	-	30.2%
Secondary school / college	27,3%	20,3%	25,4%	-	14,3%	6,0%	7,1%	62,20%	43,20%	22.7%
University or comparable	18,8%	49,0%	44,4%	85,7%	71,4%	20,0%	26,3%	29,7%	43,2%	35.5%
Other	2,6%	3,5%	2,2%			2,0%				2.1%
Paid work (more than 30 hours/week)	41,2%	51,8%	42,2%	35,7%	57,1%	18,2%	35,0%	45,9%	32,4%	40.9%
Paid work (up to 30 hours/week)	9,7%	12,2%	10,5%	7,1%	3,6%	-	-	8,1%	10,8%	8.3%
Self-employed	6,0%	4,1%	5,5%	21,4%	-	-	-	27,0%	24,3%	5.8%
Retired/pensioned	39,3%	22,8%	26,2%	28,6%	14,3%	62,6%	49,0%	2,70%	2,70%	32.5%
Not in paid work	0,7%	2,0%	2,2%	-	-	-	-	5,4%	2,7%	1.4%
Fulltime in education	1,1%	6,6%	9,5%	-	-	5,1%	4,0%	10,8%	27,0%	6.2%
Unable to work	0,4%	-	3,3%	7,1%	17,9%	4,0%	6,0%	-	-	2.5%
Other	1,5%	0,5%	0,7%	-	7,1%	10,1%	6,0%	-	-	2.4%
Living in the neighbourhood for 0–4 years	8,8%	29,8%	28,0%	-	42,9%	1,0%	5,0%	5,4%	35,1%	18.3%
Living in the neighbourhood for 5–9 years	5,9%	12,2%	19,2%	-	28,6%	-	5,0%	5,4%	2,7%	10.4%
Living in the neighbourhood for 10–20 years	10,6%	20,0%	23,1%	21,4%	14,3%	7,0%	4,0%	59,5%	13,5%	16.8%
Living in the neighbourhood for more than 20 years	74,7%	38,0%	29,7%	57,1%	14,3%	92,0%	86,0%	29,70%	48,60%	54.3%
Cabin user	b	b	b	21,4%	b	b	B	b	b	0.3%

[a] Social status was not recorded in Finland. [b] cabin use is only relevant in Driva.

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behaviours from the list they already perform. The initial list of behaviour was compiled based on a selection of individual behaviours with the highest CO_2 reduction potential as outlined in [55]. The authors then discussed internally this list and supplemented behaviours that were important for the neighbourhoods in the sample (e.g., hunting in the rural neighbourhoods). The behaviours were then grouped in four thematic areas: travel behaviour, protest behaviour, diet, and general pro-climate behaviour. For each group of behaviours a Rasch-Model [56] of a latent variable was specified to accompany for different difficulties of the behaviours (see S1 Table for the estimated item difficulties). Technically, this was done by fixing all loadings of

Table 3. Final structural equation model with all non-significant variables and relations removed. Model Fit: Chi^2 = 1631.744, df = 709, Chi^2/df = 2.30; RMSEA = .035 [.032.037]; CFI = .885; TLI = .876; SRMR = .080.

	B	SE	beta	p
Latent constructs				
Collective efficacy ← COEFF1	1.000	-	.673	-
Collective efficacy ← COEFF2	1.030	.062	.713	< .001
Collective efficacy ← COEFF3	1.018	.068	.690	< .001
Social Norms ← SN1	1.000	-	.772	-
Social Norms ← SN2	.754	.069	.578	< .001
Social Capital ← CAP1	1.000	-	.519	-
Social Capital ← CAP2	1.260	.095	.680	< .001
Social Capital ← CAP3	1.321	.114	.692	< .001
Social Capital ← CAP4	1.272	.111	.638	< .001
Pro-climate Attitude ← ATT1	1.000	-	.780	-
Pro-climate Attitude ← ATT2	1.174	.052	.822	< .001
Pro-climate Attitude ← ATT3	1.105	.073	.685	< .001
Individual Efficacy ← INDEFF1	1.000	-	.676	-
Individual Efficacy ← INDEFF2	1.143	.058	.746	< .001
Individual Efficacy ← INDEFF3	1.203	.068	.703	< .001
General Climate Behaviour ← BEH01	.497	.026	.725	< .001
General Climate Behaviour ← BEH02	.497	.026	.725	< .001
General Climate Behaviour ← BEH03	.497	.026	.725	< .001
General Climate Behaviour ← BEH04	.497	.026	.725	< .001
General Climate Behaviour ← BEH05	.497	.026	.725	< .001
General Climate Behaviour ← BEH06	.497	.026	.725	< .001
Travel Behaviour ← BEH07	.315	.035	.569	< .001
Travel Behaviour ← BEH08	.315	.035	.569	< .001
Travel Behaviour ← BEH09	.315	.035	.569	< .001
Travel Behaviour ← BEH10	.315	.035	.569	< .001
Travel Behaviour ← BEH11	.315	.035	.569	< .001
Travel Behaviour ← BEH12	.315	.035	.569	< .001
Travel Behaviour ← BEH13	.315	.035	.569	< .001
Protest Behaviour ← BEH14	.610	.034	.844	< .001
Protest Behaviour ← BEH15	.610	.034	.844	< .001
Protest Behaviour ← BEH16	.610	.034	.844	< .001
Second order latent construct				
Climate Action ← General Climate Behaviour	1.000	-	.763	-
Climate Action ← Protest Behaviour	.940	.148	.743	< .001
Climate Action ← Travel Behaviour	1.355	.270	.848	< .001
Climate Action ← Diet	.597	.073	.649	< .001
Structural paths				
Individual intention → Climate Action	.254	.054	.268	< .001
Collective intention → Climate Action	.087	.040	.082	.030
Pro-climate Attitude → Climate Action	.372	.084	.223	< .001
Social Capital → Climate Action	-.200	.095	-.090	.034
Household Size → Climate Action	.089	.033	.122	.007
Children<14 → Climate Action	-.206	.082	-.136	.012
University Degree → Climate Action	.244	.085	.098	.004
Finland → Climate Action	-1.055	.161	-.347	< .001

(*Continued*)

Table 3. (Continued)

	B	SE	beta	p
Individual Efficacy → Individual Intention	1.429	.102	.730	< .001
Pro-climate Attitude → Individual Intention	-.230	.083	-.131	.005
University Degree → Individual Intention	.250	.078	.096	.001
Male → Individual Intention	-.291	.087	-.117	.001
Finland → Individual Intention	-1.043	.127	-.325	< .001
Social Capital → Collective Intention	.613	.086	.295	< .001
Social Norms → Collective Intention	.284	.067	.215	< .001
Collective Efficacy → Collective Intention	.160	.051	.107	.002
Household Size → Collective Intention	-.062	.028	-.089	.028
Children<14 → Collective Intention	.245	.068	.173	< .001
Working → Collective Intention	-.293	.073	-.132	< .001
Finland → Collective Intention	-.922	.095	-.323	< .001
Male → Pro-climate Attitude	-.246	.049	-.173	< .001
Finland → Pro-climate Attitude	-.567	.061	-.310	< .001
Finland → Individual Efficacy	-.469	.070	-.286	< .001
Covariations				
Pro-environmental Attitudes ←→ Collective Efficacy	.321	.029	.656	< .001
Pro-environmental Attitudes ←→ Social Norms	.135	.024	.244	< .001
Pro-environmental Attitudes ←→ Social Capital	.071	.016	.202	< .001
Pro-environmental Attitudes ←→ Individual Efficacy	.262	.022	.652	< .001
Individual Efficacy ←→ Collective Efficacy	.420	.035	.928	< .001
Individual Efficacy ←→ Social Norms	.202	.025	.396	< .001
Individual Efficacy ←→ Social Capital	.036	.013	.110	.005
Social Capital ←→ Collective Efficacy	.106	.019	.268	< .001
Social Capital ←→ Social Norms	.258	.029	.577	< .001
Social Norms ←→ Collective Efficacy	.305	.035	.490	< .001
INDEFF3 ←→ COEFF1	.151	.021	.238	< .001

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the items of one domain to the same value and fixing the variance of the corresponding latent variable to 1. Since the diet question were overlapping, answers to the three questions were aggregated as follows: People who reported eating vegan, were scored 3 in diet, vegetarians were scored 2, people on a diet low on animal products were scored 1, and all others were scored 0. Consequently, diet was not included in the Rasch modelling. Fig 2 shows the percentage of people performing the behaviour for all included behaviours, colour-coded by domain, a corresponding figure with item difficulties can be found in S1 Fig.

Individual intention and collective intention. *Individual* and *collective intentions* were measured by one item each, following the standard intention items as used in the Theory of Planned Behaviour research [57], but adapted to the individual and collective focus (see S2 Table for a complete list of items).

Individual level predictors. On the individual level, attitudes, perceived individual efficacy, and social norms were measured as recommended in Theory of Planned Behaviour research. *Attitudes* were measured by three items, which all loaded strongly on one factor. Cronbach's alpha of the composite score was.81. *Individual efficacy* was also measured by three items, which all loaded strongly on one factor (Cronbach's alpha.77).

Collective level predictors. *Perceived collective efficacy* was measured by three items adapted from Hamann, Holz, and Reese [58]. The items load strongly on one factor

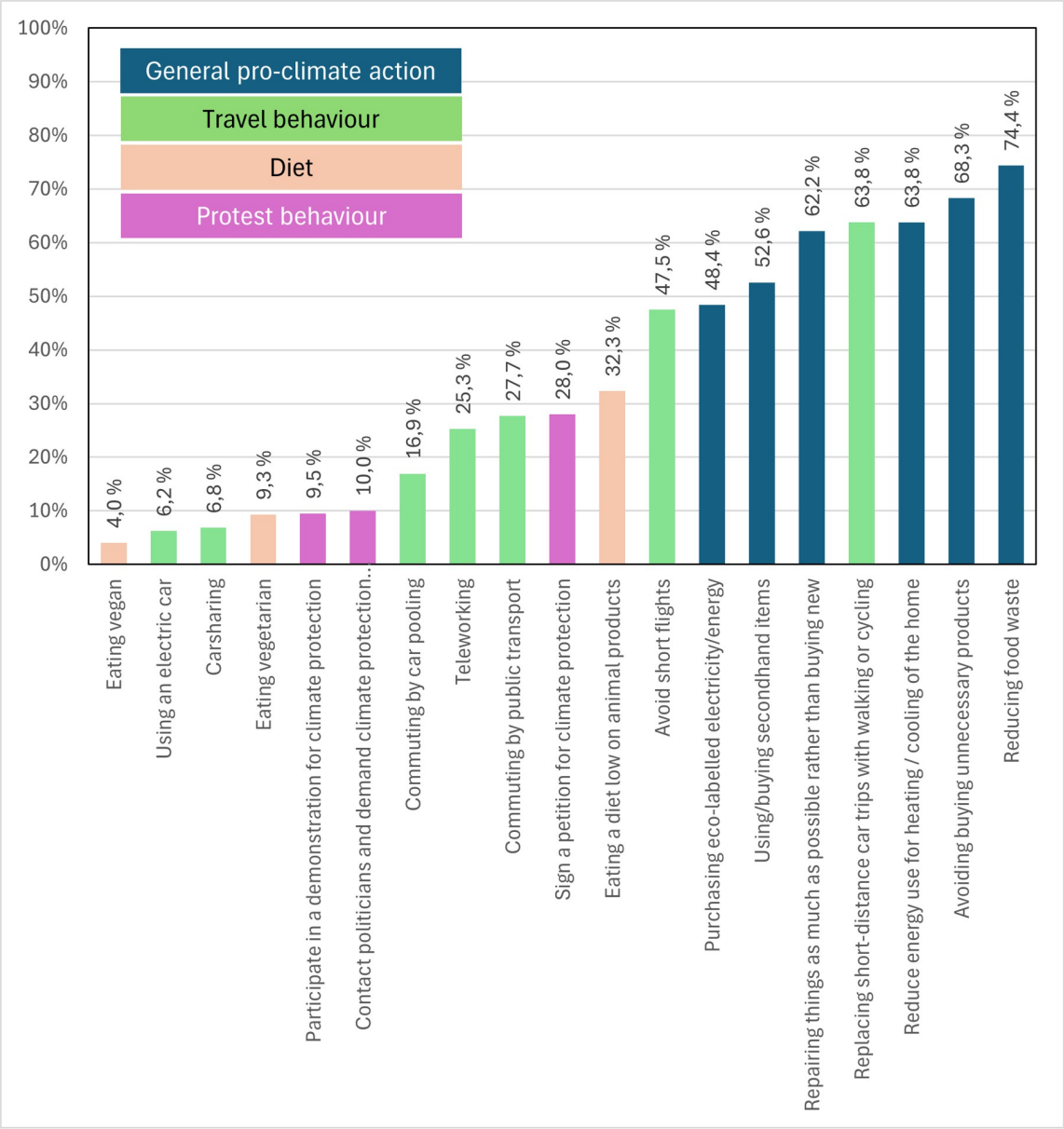


Fig 2. Percentage of people self-reporting to implement each of the tested behaviours, color-coded by behavioural domain. Please see S1 Fig for a corresponding figure with item difficulties.

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(Cronbach's alpha.74). All four items have strong loadings on one factor (Cronbach's alpha.92). Social capital in the neighbourhood was measured by four items capturing components developed specifically for this study. Loadings on one factor are medium size to strong (Cronbach's alpha.73). Finally, *social norms* were measured with two items, with strong loadings on one factor (Cronbach's alpha.63).

Cultural level predictors. On the level of local culture, two proxy variables have been used: One item indicating the country of residence to capture the national cultural differences and one item indicating if the neighbourhood was rural or (semi)urban to capture the dimension of rural vs. urban lifestyles. Country was initially dummy coded with Austria (the largest

sub-sample) as a reference category. However, no country differences showed but for Finland. Thus, Finland was tested against all other countries combined.

Structural predictors. On the structural level, the socio-demographic variables as listed in Table 2 were recorded (with the categories listed there). Social status was measured by an item which was used in previous research [52]. The respondents were asked to indicate where they consider themselves on a ladder where the lowest step (1) represents the people that are the worst off and the highest step (10) represents the people that have it exceptionally well. As this measure was not implemented in the Finnish sample, it was not included in the structural equation model.

For the analyses, gender was recoded into a dummy variable with identification as male as 1 and all other categories as 0. Education was dummy coded into having a university degree (1) vs. all other categories (0). The job situation was dummy coded into working (full time, part time or self-employed = 1) vs. all other categories (0).

Results

Structure and difficulties of climate behaviours

As Fig 2 above shows clearly, the frequency with which our participants report performing climate actions differs greatly between the behaviours and also a shows a pattern regarding the four domains: General climate action as in reducing food waste, reducing purchases, using eco-labelled energy, reducing indoor temperature or repairing and buying second hand are—according to self-reports—already done by many, if not most people, two travel related behaviours fall also in this category: Avoiding short flights and using active modes for short-distance travel. All other travel behaviours, but also diet and protest behaviours are less common. Especially, more active protest behaviours, plant-based diets, and electric cars and car sharing are not common.

Structural model of neighbourhood climate action

Fig 3 and Table 3 show the final structural equation model after removal of insignificant paths and the addition of three paths based on modification indices (see method section). According to most model fit indices this model has an acceptable to good fit to the data: The Chi^2 to df ratio is 2.30 and with that clearly below the recommended 3–5 that indicates an acceptable Chi^2 difference. RMSEA indicates a good fit, SRMR indicates an acceptable model fit, CFI and TLI are close to, but lower than the cut-off point for an acceptable model fit. As the model is relatively complex in relation to the number of participants, we treat these last to results as tolerable.

As can be seen in Fig 3, all indicators load well on their respective latent constructs and there are no cross-loadings specified. In the second order latent variable, it can be seen that all four components contribute substantially to the general climate action with travel behaviour being the strongest second order indicator and diet the weakest (but still substantial).

Individual and collective intention both impact the degree of climate action taken positively, with individual intentions clearly being the stronger influence. In addition, attitudes have a positive direct influence on climate action (but negative on individual intention), whereas social capital has a negative direct impact on climate actions (but a positive on collective intention). Both these effects are likely suppressor effects (for which the WLSMV estimator is known to produce sometimes), which means that an "overestimated" positive effect on one variable is compensated by a negative effect of the same variable on another connected variable. Participants from the two Finnish neighbourhoods report lower climate action than participants from the other neighbourhoods. Participants with a university degree and in larger households report more climate actions, whereas the number of children below 14 years

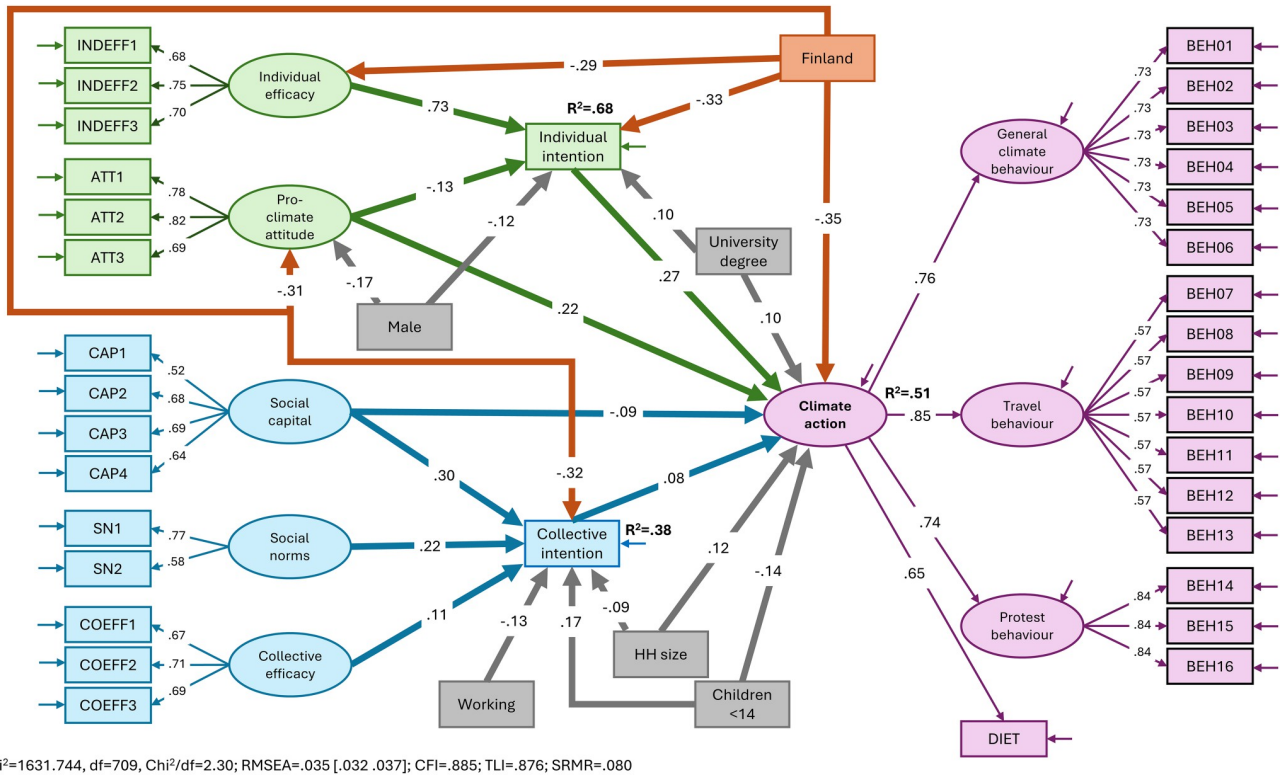


Fig 3. The final version of the multi-perspective model (all predictors included in the figure are significant, non-significant relations and variables not related to any have been removed from the model; the numbers represent standardized regression weights from the full models and adjusted explained variances).

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has a negative influence. Also, for household size and number of children, a possible suppressor effect occurred, as opposite effects of these variables were found on collective intentions.

Individual intention is mainly determined by individual efficacy, attitudes were only found to have a small and even negative effect (but see comment on suppressor effect above). The positive influence of pro-climate attitudes was rather found to be direct on climate actions. People in the Finnish neighbourhoods reported weaker individual intentions, weaker individual efficacy, and weaker attitudes. Males also reported weaker individual intentions to act and weaker pro-climate attitudes, whereas a university degree was positively related to stronger individual intentions.

Collective intentions were positively influenced by the social capital in the neighbourhood, social norms to act, and collective efficacy. The effect of social capital seems to be mostly mediated through collective intentions. Again, participants from the two Finnish neighbourhoods report a weaker collective intention. Working full time and larger households reduce the intention to act collectively in the neighbourhood, whereas the number of children above 14 increases collective intentions (but see comment on suppressor effects above).

Table 4 below summarizes the findings in relation to the hypotheses formulated in the introduction.

Discussion

Our study was one of the first analysing drivers and barriers of local climate action in neighbourhoods quantitatively from a multi-perspective approach. Based on extensive qualitative

Table 4. Confirmation of hypotheses.

Hypothesis		Confirmed?
H1a	Individual intentions to act against climate change result in a higher level of (self-reported) climate action.	Confirmed
H1b	Higher perceived individual efficacy results in stronger individual intentions to act against climate change.	Confirmed
H1c	Stronger pro-climate attitudes result in stronger individual intentions to act against climate change.	Not confirmed, but positive direct effect on climate actions
H2a	Collective intentions to act against climate change on the neighbourhood level result in a higher level of (self-reported) climate action.	Confirmed
H2b	Higher perceived collective efficacy results in stronger collective intentions to act against climate change.	Confirmed
H2c	Stronger social norms result in stronger collective intentions to act against climate change.	Confirmed
H2d	Higher social capital in the neighbourhood results in stronger collective intentions to act against climate change.	Confirmed
H3a	There are country differences in central variables like individual and collective intentions and climate action, as well as in the predictors of these. The direction of the impact is not determined a priori.	Confirmed
H3b	There are differences between rural and urban neighbourhoods in central variables like individual and collective intentions and climate action, as well as in the predictors of these. The direction of the impact is not determined a priori.	Not confirmed, no significant effects removed from model
H4a	Age has an influence on central variables like individual and collective intentions and climate action, as well as predictors of these. The direction of the impact is not determined a priori.	Not confirmed, no significant effects removed from model
H4b	Participants who identify as males score lower on central variables like individual and collective intentions and climate action, as well as predictors of these.	Confirmed
H4c	Increased household size leads to lower scores on central variables like individual and collective intentions and climate action, as well as predictors of these.	Partly confirmed
H4d	Number of children bellow 14 years of age leads to higher scores on central variables like individual and collective intentions and climate action, as well as predictors of these.	Partly confirmed
H4e	Higher education leads to higher scores on central variables like individual and collective intentions and climate action, as well as predictors of these.	Confirmed
H4f	Working full time leads to lower scores on central variables like individual and collective intentions and climate action, as well as predictors of these.	Confirmed

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research in nine neighbourhoods and a literature review, we designed a questionnaire which covered potential drivers from the individual, the collective, the broader cultural, and the socio-structural level. Potentially the most relevant finding is that, indeed, factors from all four levels were shown to be relevant for local climate action. This shows clearly that focussing on individual drivers alone will necessarily ignore relevant boundary conditions of action, but also that ignoring the individual level factors falls equally short. As such, the study also outlines a way to study the interplay of such different factors also for future work.

In general, it appears that, if studied with a survey on residents living in a neighbourhood, individual intentions to act against climate change clearly outweigh the impact of collective intentions to act. Thus, for most people we studied, climate change seems not to be a collective issue, but rather a private issue. This is also confirmed by the preceding qualitative work, where we learned that climate change was usually not a topic discussed a lot in the

neighbourhoods (see short summaries of that research in the S3 Table). However, drawing on results from previous studies on collective action [15, 59], one may anticipate that the capacity and motivation to act against climate change in a neighbourhood could be strengthened if it were framed more as a collective topic. We see the significant but small effect of collective intentions rather as a potential for strengthening future collective action than as a barrier to collective action.

Confirming findings of previous research [30, 32], we find that individual attitudes and individual efficacy are relevant individual-level factors that could stimulate a strong individual intention to act. Surprisingly, pro-climate attitudes showed a negative effect on individual intentions, but a positive direct effect on climate action, which—likely being a suppressor effect—indicates that the main influence seems to be going directly and that a lot of the individual intention is actually determined by perceived individual efficacy. In our case, perceived individual efficacy is by far the strongest factor, which underlines that strengthening people's belief in the efficacy of their actions is probably the most important strategy to stimulate their willingness to act on an individual level. From a neighbourhood perspective, it is also important to understand that a feeling of collective efficacy has been shown to increase pro-environmental intentions through a positive effect on individual efficacy [60], in other words, believing to be able to make a difference together with other people increases also one's own perceived efficacy.

Interestingly, when including collective intentions into the analysis, social norms no longer have an effect on individual intentions, as predicted in the theory of planned behaviour [31], but on the collective intention. This makes sense from a theoretical perspective: If I experience social pressure to act against climate change, this would then rather create an intention to act together with others who excerpt this pressure than alone. Neighbourhood social capital—hence the ability of a neighbourhood to work together creatively and efficiently to face challenges—has the strongest impact on collective intentions. This replicates findings from other research on the role of social capital in local adaptation to climate change [38, 61, 62]. It shows that social capital is not only decisive in collective action to protect a neighbourhood against the impacts of climate change, but also in mitigative action.

When zooming in more on the specific results for culture and socio-structural impacts, which are—due to their contextuality—likely not generalizable to all neighbourhoods, we find further interesting effects: We see consistently that in the Finnish neighbourhoods, collective and individual intentions to act against climate change as well as the actions taken are weaker than in the other three countries. This is surprising, given that in a comparative European analysis, Finnish respondents were about the same level concerned about climate change as Norwegian, Austrian, and Italian respondents [63]. However, in the same study it was found that Finnish respondents expected less negative outcomes of climate change as compared to the other countries in our study, which might explain this effect. Furthermore, the two Finnish neighbourhoods were the by far most rural neighbourhoods, where the participants often lived far apart from each other, even if counted as one neighbourhood. This might also mean that for these people conditions are so different "deep in the forest", that they feel less need to act against climate change and rather focus on other challenges (like transport).

Finally, we found that the following socio-structural components were relevant for taking climate actions and individual or collective intentions. In line with other studies that show a larger group of male respondents being less concerned about climate change in western societies [52, 64], we also find that respondents identifying as male report weaker pro-climate attitudes and lower individual intentions to act, but interestingly not weaker climate action. High education contributed positively to individual intentions to act and climate actions, which again aligns with other research [65]. Household size was in our study positively related to

climate actions, but this was counter-balanced by the number of family members below 14 years, which has a negative impact on the number of climate action. This contradicts [52], who find that the number of household members was negatively related to climate action and the number of children was irrelevant. Maybe, part of the explanation lies in that the number of children positively affects the collective intention, possibly through more interaction with other families in the neighbourhood that young families have. Working is negatively related to collective intentions to act, possibly an expression of less time spent on the neighbourhood.

Taken together, our results have some implications for policy and planning: First of all, most of the climate action is (still) motivated on the individual level, and here mostly by perceived individual efficacy. Strengthening this component is therefore a clear recommendation, for example by making the relevance and not the least also the positive results of household climate action salient for people. On the other hand, there lies substantial potential in framing climate action more as a collective action to strengthen the effect of collective intentions. Working together and achieving a positive effect not only motivates through stronger social norms, but also strengthens collective and (indirectly) individual efficacy. Creating arenas for such joined actions should thus be a priority by (a) picking up on grassroot initiatives that already implement these actions, or (b) by initiating such actions liaising with local key actors. A key component for the success of these collective actions seems to be high social capital in the neighbourhood. As neighbourhood social capital is characterized by the capacity to bond with neighbours, to "come out and talk" [66] measures to strengthen social capital can be manifold, but might for example include to restructure neighbourhoods physically to create arenas where neighbours can meet and talk. An excellent example is the restructuring of car-dominated neighbourhoods in Barcelona into public spaces for meeting other people and being outside (the so-called superblocks) [67].

Limitations

While this study is producing interesting and relevant results, it has also a number of limitations that need to be acknowledged and kept in mind when interpreting the findings. First, the study is based on a relatively small number of neighbourhoods (9) from only four countries, which means that the effects on the cultural and socio-structural level are likely very contextual and cannot be quantified. In a larger study, it would be ideal to survey people in many neighbourhoods in many countries to be able to also quantify variation on this level in a statistical multilevel analysis. Second, the neighbourhoods involved are all located in Europe (even if they span from Northern to Southern Europe). To get a more comprehensive understanding of the global potential of neighbourhood action against climate change, a similar methodology should be implemented also in non-European countries, also offering a larger variability of cultural contexts. Third, the number of people surveyed in the neighbourhoods was very different for the nine neighbourhoods, resulting in people from different neighbourhoods being unevenly represented in the results. This had also implications for the statistical analyses possible, as for example a formal test of measurement invariance across the neighbourhoods was not possible. Fourth, the cultural dimensions country and urban vs. rural are not independent of each other as both Finnish neighbourhoods were the most rural. This means that Finland and rural are confounded in the analyses, which might explain, why the rural-urban dimension did not contribute significantly. Sixth, we focused exclusively on the positive effects of neighbourhoods on local climate action, but there might also be negative, when frontrunners of climate friendly lifestyles clash with their neighbours about what is acceptable (for example when solar panels are installed on some houses).

Conclusions

This quantitative multi-neighbourhood study on collective climate action demonstrated that factors from the individual, collective, cultural, and socio-structural level together determine if people engage in actions to mitigate climate change. This underlines the importance of focussing on all these levels, when trying to stimulate engagement. Currently, local climate action seems to be framed as an individual behaviour, rooted mostly in individual level drivers. However, the analyses also indicate that there is a large potential for increased engagement especially in neighbourhoods with strong social capital, if social norms support climate action. Cultural differences shape how the different factors influence intentions and actions, which underlines the need to tailor intervention approaches to local cultures.

Supporting information

S1 Table. List of behaviours the respondents could chose the ones they already implement (including the percentage of people who said that they do so). Item difficulty is estimated in the Rasch-model.
(DOCX)

S2 Table. Item lists for the included variables. Measured on 5-point Likert-type scales (e.g., 1 = totally disagree, 5 = totally unless indicated otherwise).
(DOCX)

S3 Table. Short summary of the results of the qualitative analyses in the neighbourhoods, based on the project report (Klöckner et al., 2023, pages 27–49). The analysis are based on analyses of a total of 41 official documents from the municipalities provided for the neighbourhoods, 35 interviews with key stakeholders, and interviews with 51 residents from the neighbourhoods.
(DOCX)

S1 Fig. Visualiztion of item difficulties as estimated in the Rasch-model.
(TIF)

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